EECS 482
Introduction to operating systems

Dynamic address translation

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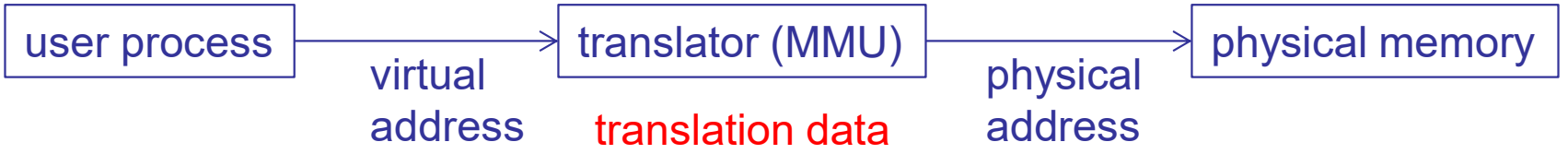
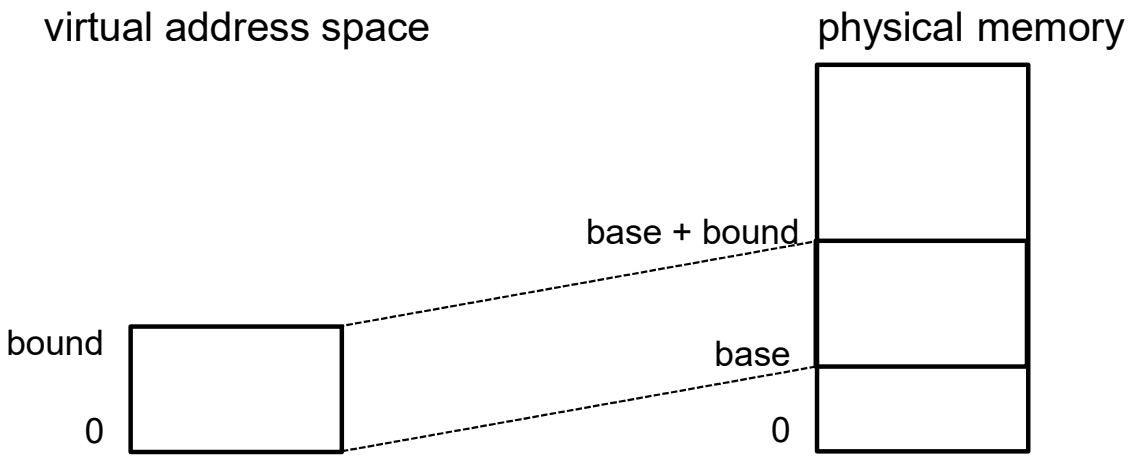
Dynamic address translation



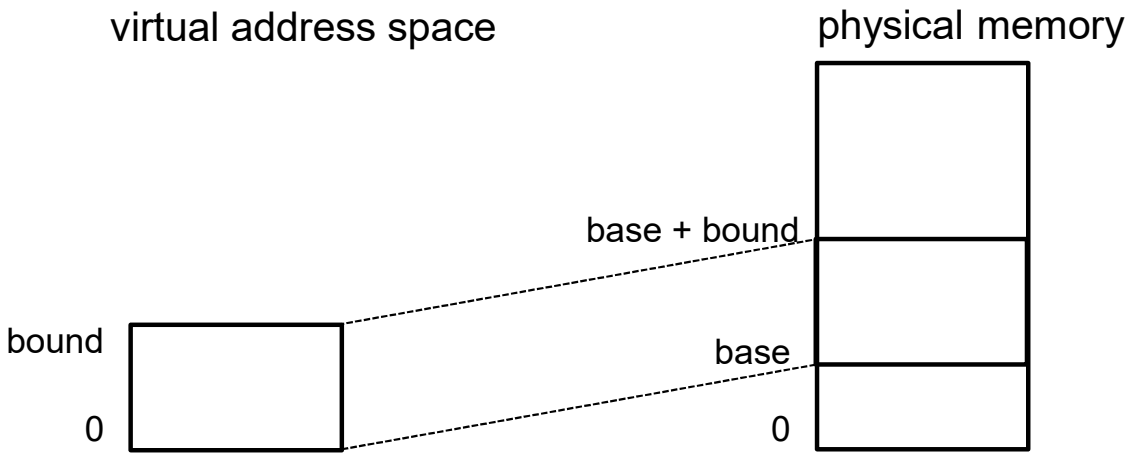
- Translator typically implemented as part of CPU
- Translator uses translation data
  - Each process has its own translation data
  - Changing address spaces = changing data used to translate a virtual address
- Many ways (data structures) to implement translator
  - Speed of translation
  - Size of data needed to support translation
  - Flexibility (sharing, growth, large address space)

Base and bound

- Load each process into a single contiguous region of physical memory
  - Base register: starting physical address
  - Bound register: size of region



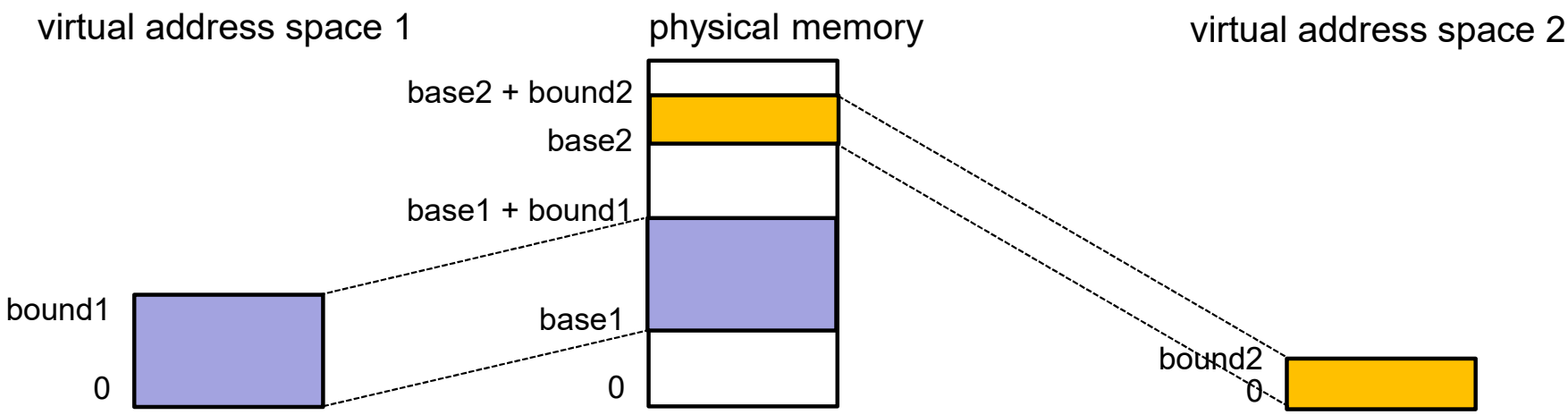
Base and bound: MMU translation algorithm




```
if (virtual address >= bound) {
        trap to kernel; OS kills process (segmentation violation)
                        or retries access
} else {
        physical address = virtual address + base
}
```

How to switch address spaces?

Base and bound: switching address spaces



Base and bound

- Fast
- Simple hardware support

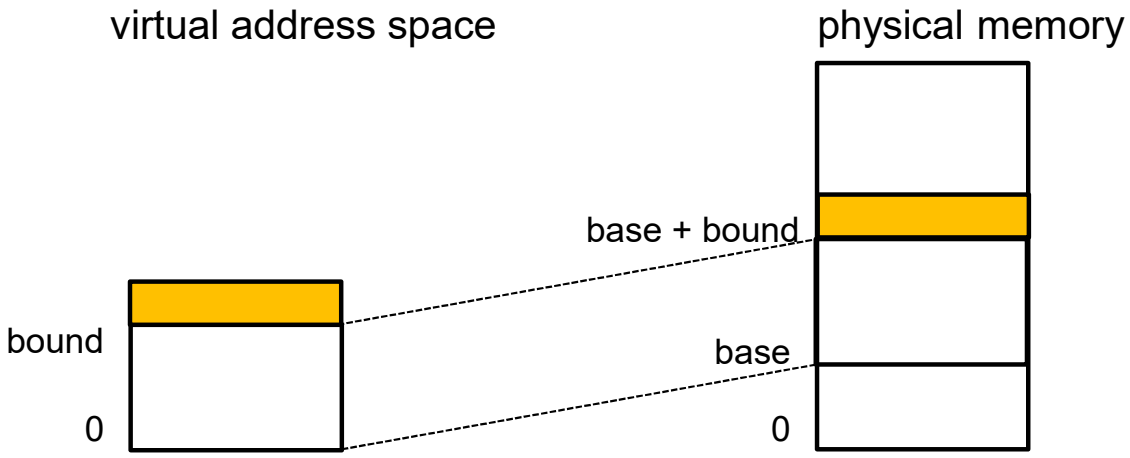
✓ Address independence?

✓ Protection?

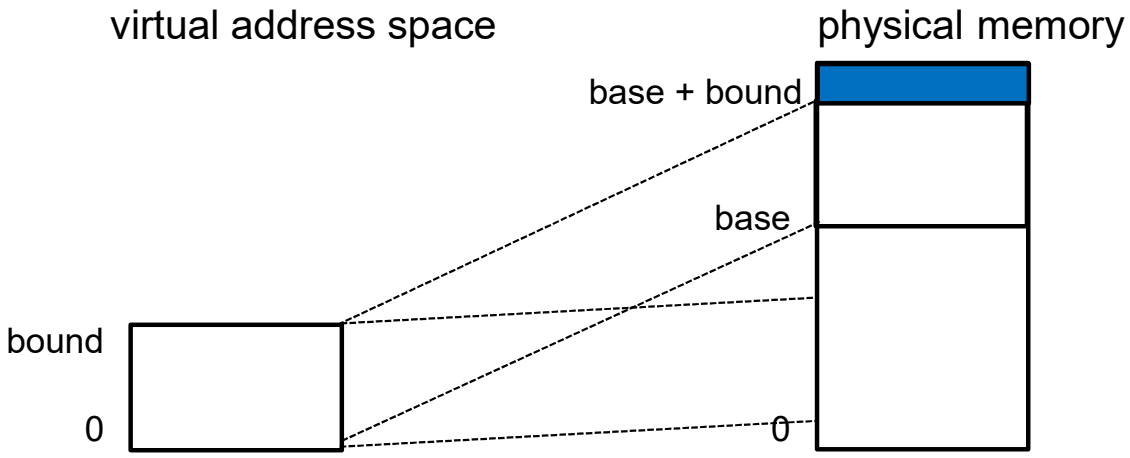
✘ Large address space?

- Flexibility?

Base and bound: how to grow address space?



Base and bound: how to grow address space?



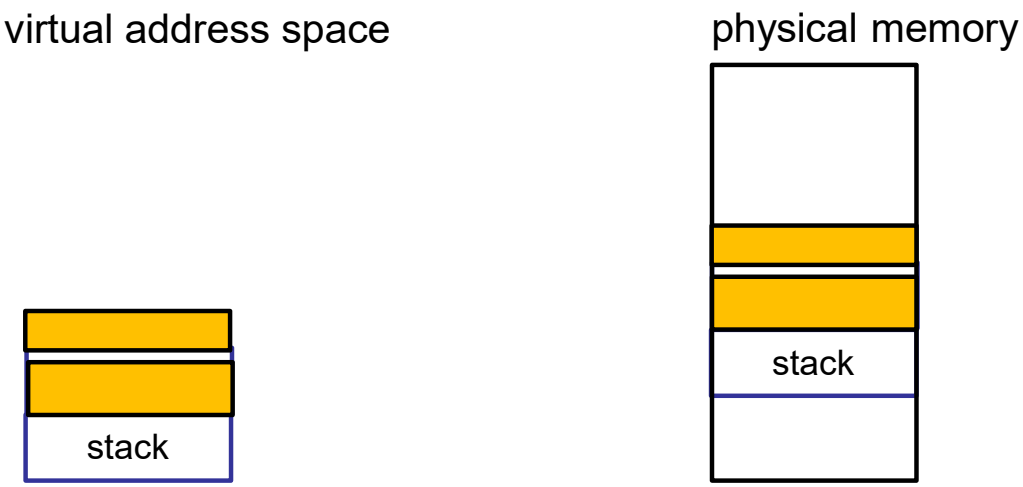
Is process aware of this move?

+ Transparent

- Slow

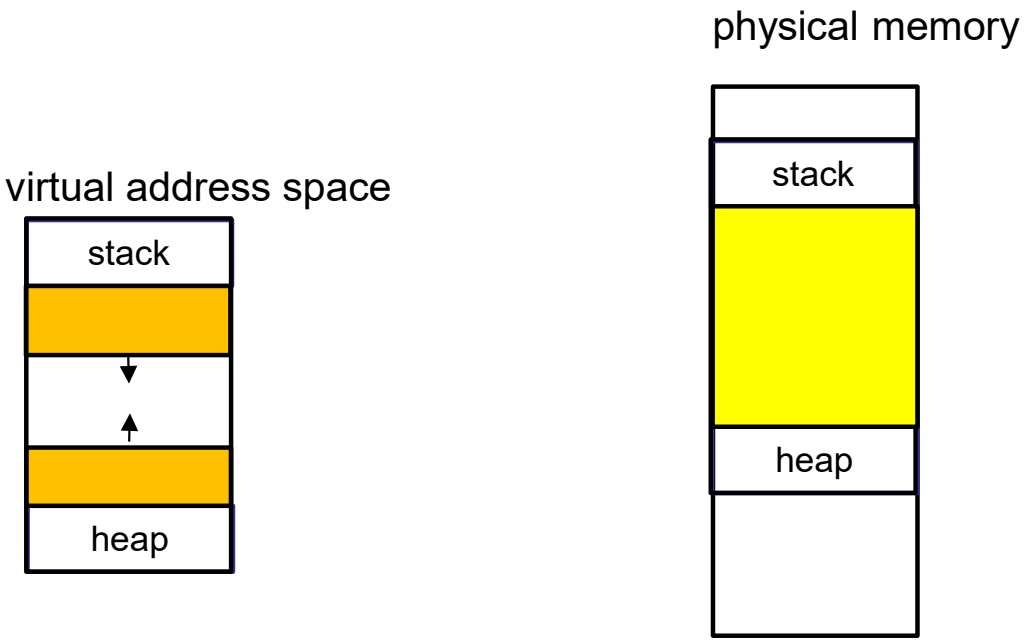
Base and bound: how to grow multiple regions in address space?

- Easy to grow top region (e.g., heap)
- How to grow lower region (e.g., stack)?
  - Problem?



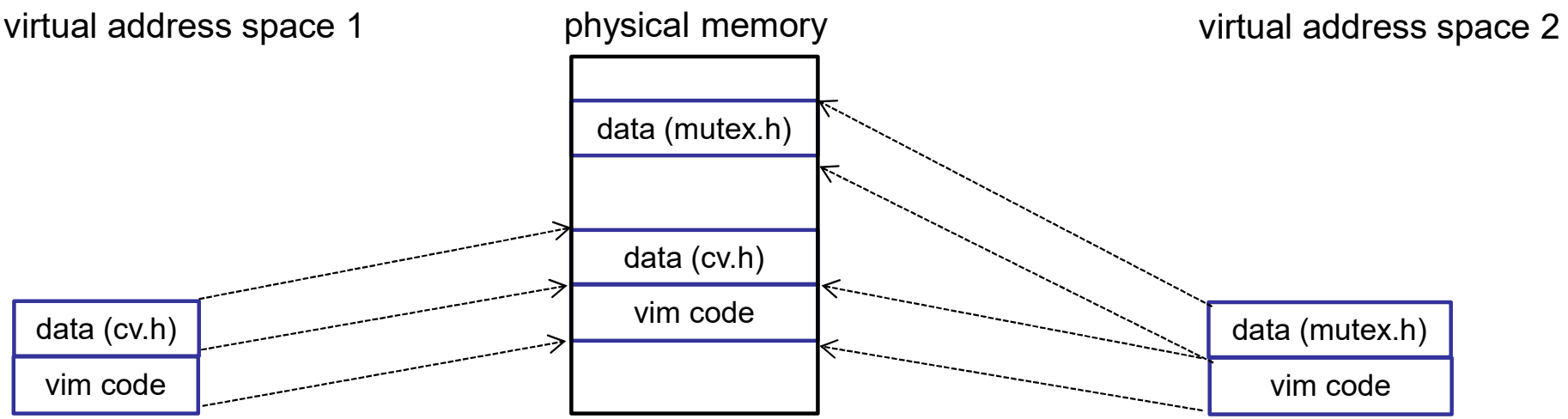
Base and bound: how to grow multiple regions in address space?

- Must reserve space for each region in virtual address space
- Problem?



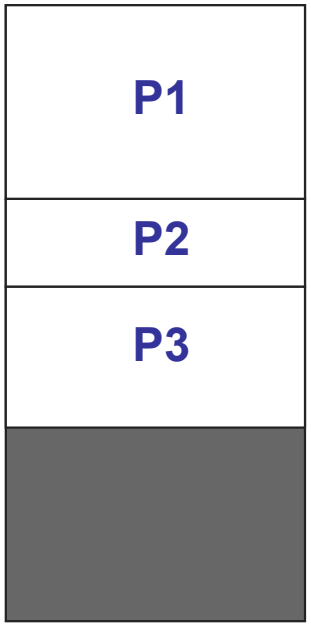
Base and bound: sharing

- Can’t share part of an address space between processes



External fragmentation

- Processes come and go, leaving a mishmash of available memory regions
  - Wasted memory between allocated regions



Base and bound

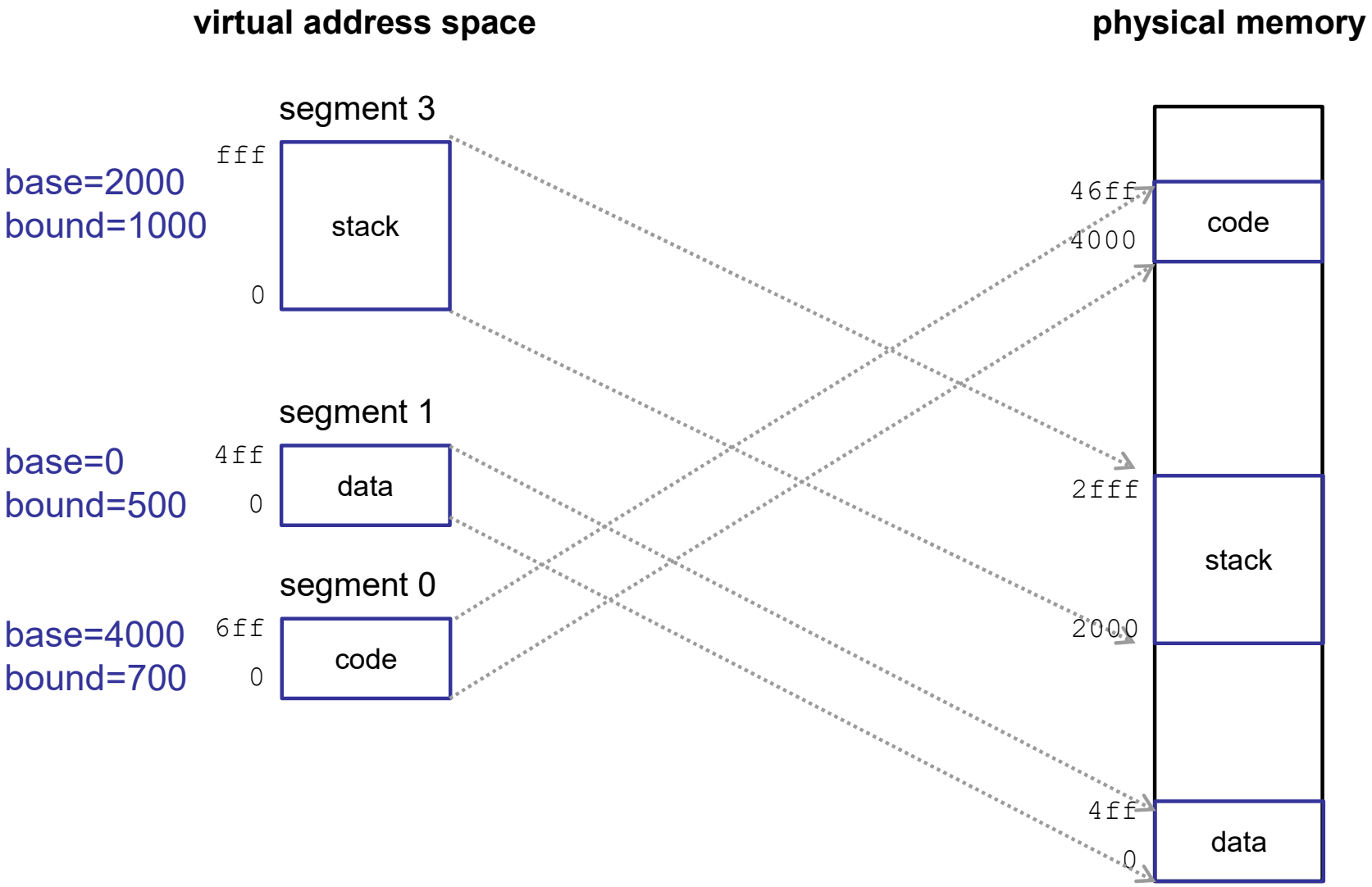
- Pros:
  - Fast
  - Simple hardware support
- Cons:
  - Does not support large address spaces
  - Growing address space may be slow
  - Need to reserve space to allow multiple regions to grow
  - Can’t share part of address space
  - External fragmentation
- Root cause: requires entire address space to be contiguous in memory

Segmentation

- Divide address space into multiple segments

- Segment: region of address space that is:
  - Contiguous in physical memory
  - Contiguous in virtual address space
  - Variable size

Segmentation



Segmentation

Segment #	Base	Bound	Description
3	2000	1000	stack segment
2	n/a	0	unused
1	0	500	data segment
0	4000	700	code segment

- Virtual address: (segment #, offset)
  - Physical address = seg_info[segment #].base + offset
- How to specify the segment #:
  - High bits of address
  - Special register
  - Implicit to instruction opcode

Segmentation: translation

Segment #	Base	Bound	Description
3	2000	1000	stack segment
2	n/a	0	unused
1	0	500	data segment
0	4000	700	code segment

- Physical address for virtual address `(3, 100)` ?
  - `2000 + 100 = 2100`
- Physical address for virtual address `(0, ff)` ?
  - `4000 + ff = 40ff`
- Physical address for virtual address `(1, 2000)` ?
  - invalid (illegal) ➔ segmentation violation/core dump
- Physical address for virtual address `(2, ff)` ?
  - invalid (illegal) ➔ segmentation violation/core dump

Valid vs. invalid addresses

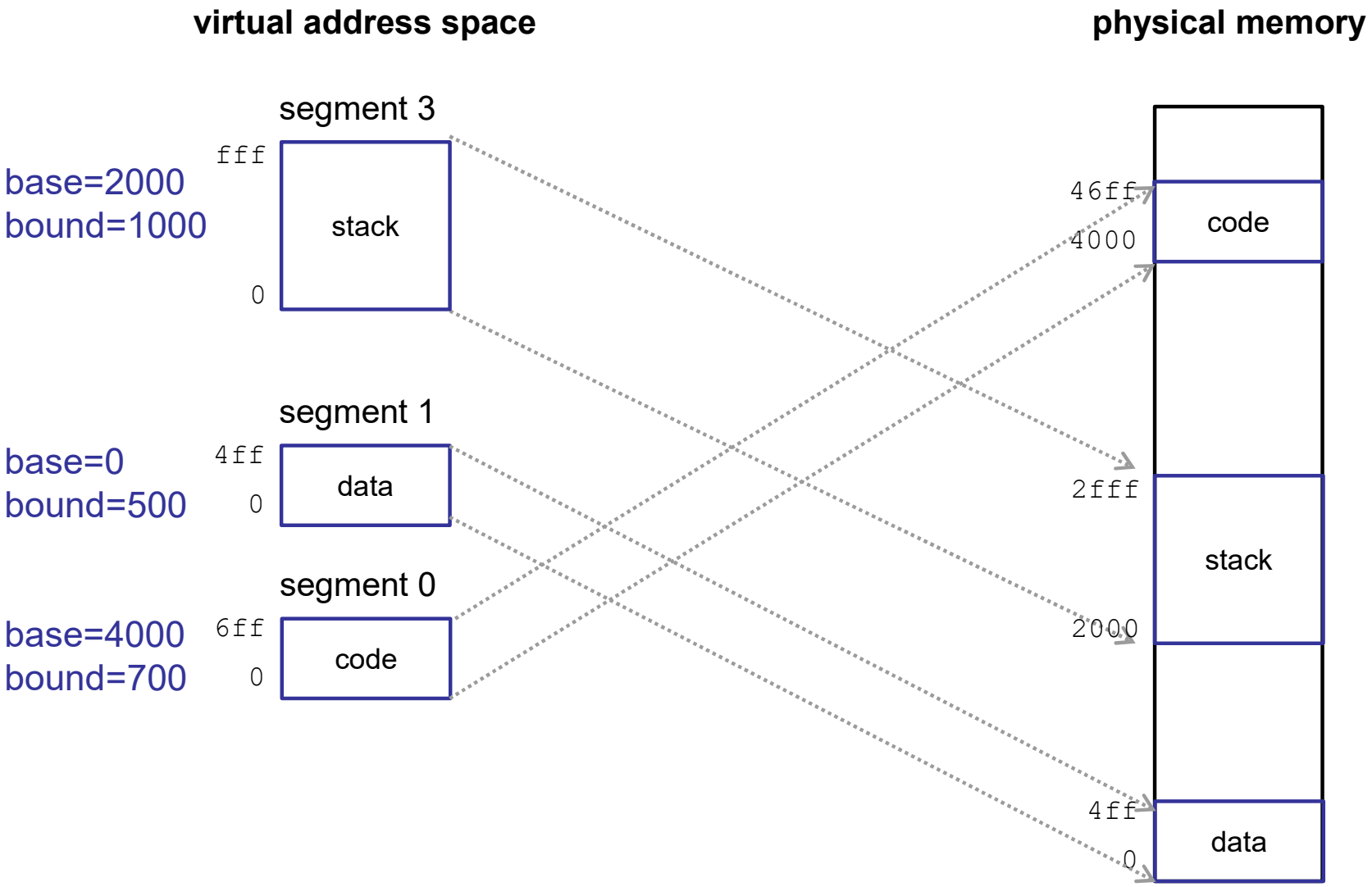
- Not all virtual addresses are valid
  - Valid → address is part of virtual address space
  - Invalid → virtual address is illegal to access
    - » Accessing invalid address causes trap to OS

- In segmentation, a virtual address can be invalid due to:
  - Out of bound offset
  - Invalid segment number

Segmentation: protection

Segment #	Base	Bound	Description	Protection
3	2000	1000	stack segment	read/write
2	n/a	0	unused	
1	0	500	data segment	read/write
0	4000	700	code segment	read

Segmentation

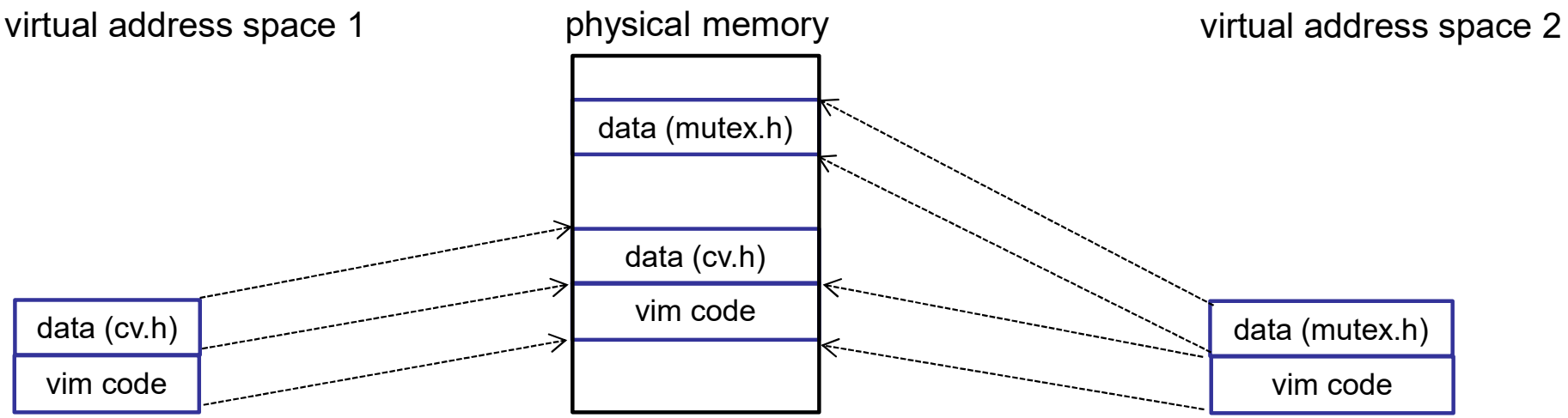


Segmentation: translation algorithm

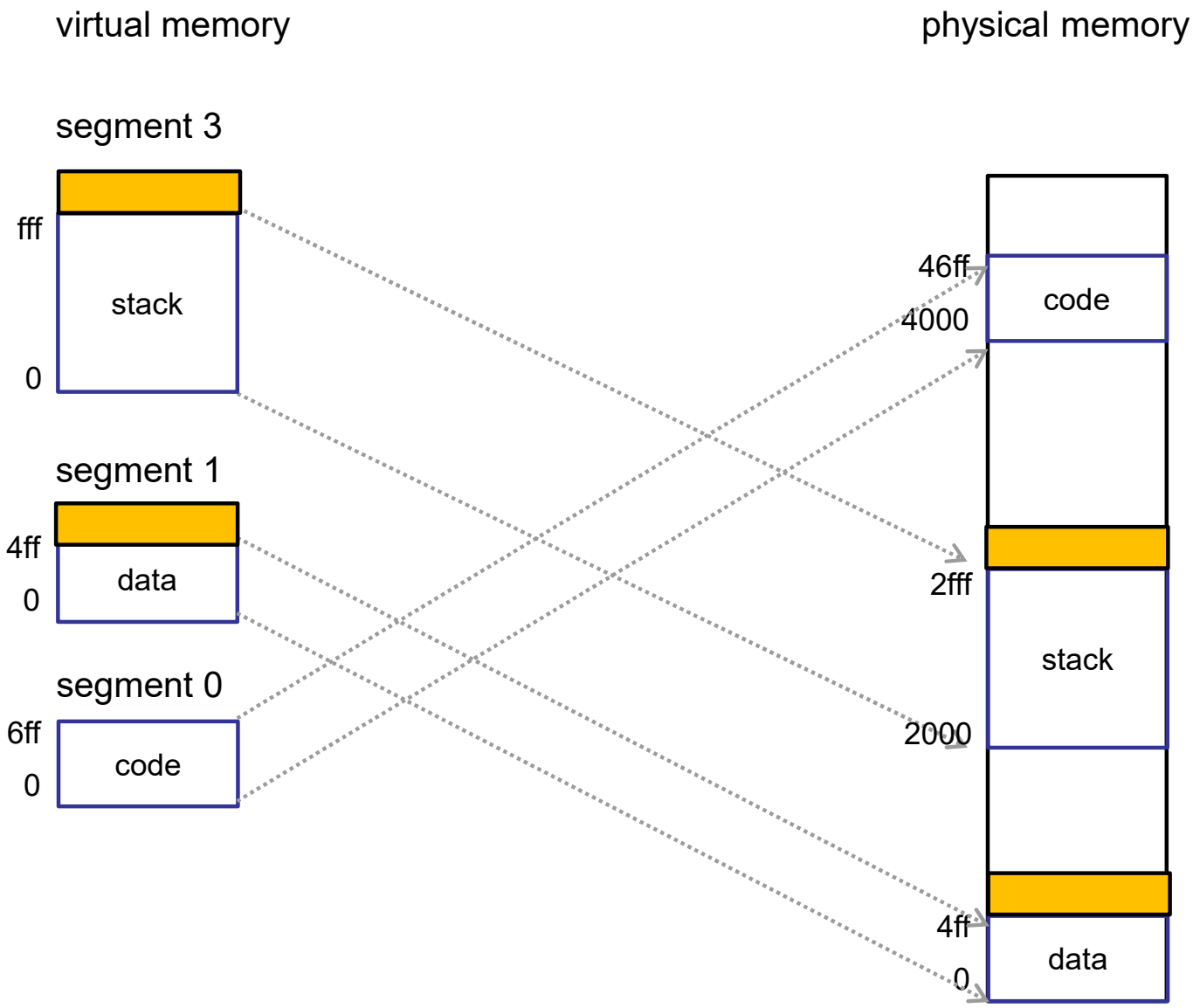
```
(base, bound, protection) = segment_info[segment]

if (segment is invalid or protected, or offset >= bound) {
        trap to kernel
} else {
        physical address = offset + base
}
```

- What must be changed to switch address spaces?

Segmentation supports partial sharing



Segmentation allows multiple growing regions

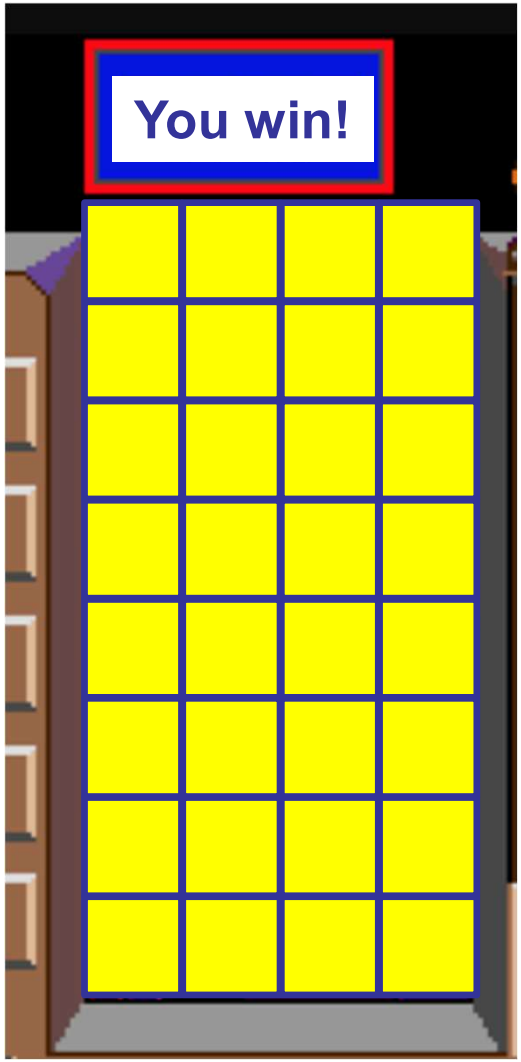


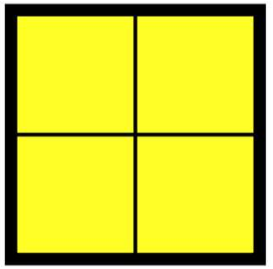
Segmentation: problems

- Growing address space still slow (may need to copy segment)
- External fragmentation
- Does not support large address spaces?
  - Each segment must fit in memory

- How can we:
  - Make memory allocation easy
  - Not have to worry about external fragmentation
  - Allow address space to be larger than physical memory

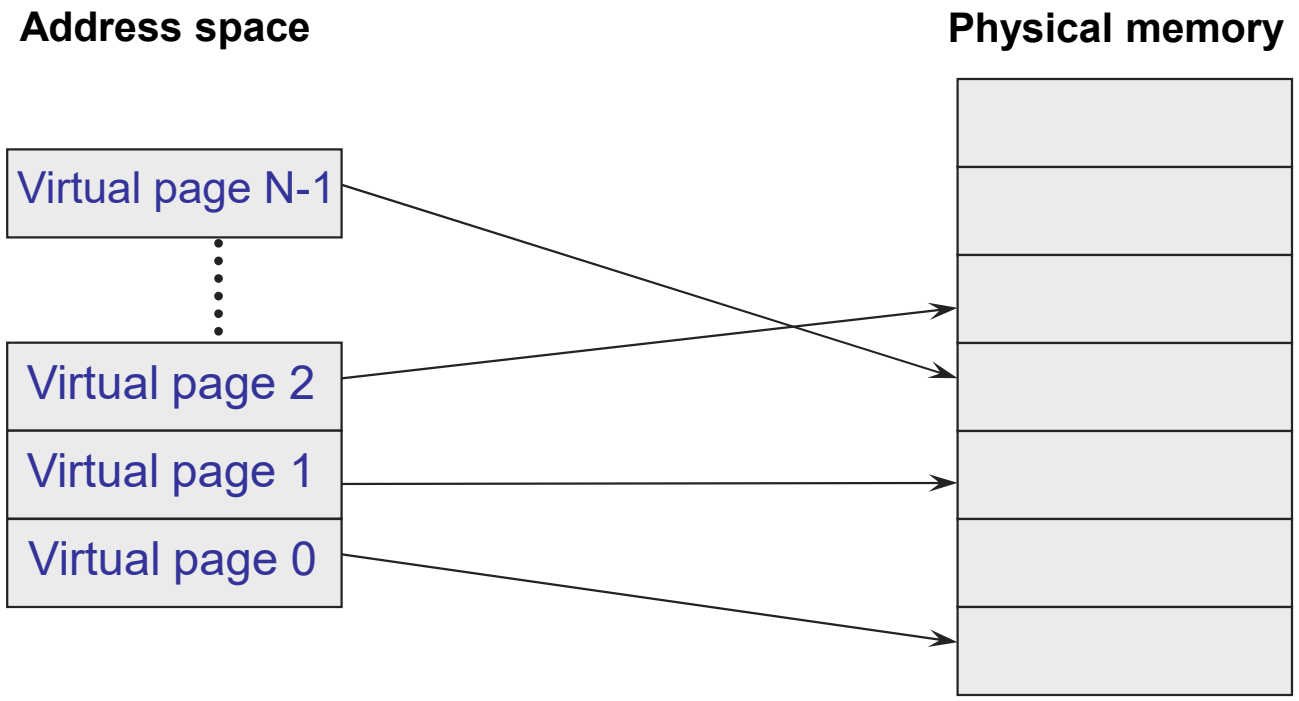
Tetris analogy





Paging

- Allocate memory in fixed-size units (pages)
  - Any free physical page can store any virtual page



Segmentation ➔ paging

Virtual page # **Physical page #**

~~Segment #~~	Base / Page Size	Bound	Valid
3	4~~000~~	~~700~~ 1000	1
2	n/a	~~0~~ 0	0
1	1b~~000~~	~~2000~~ 1000	1
0	2~~000~~	1000	1

Page ~~Segment~~ table

Segmentation ➔ paging

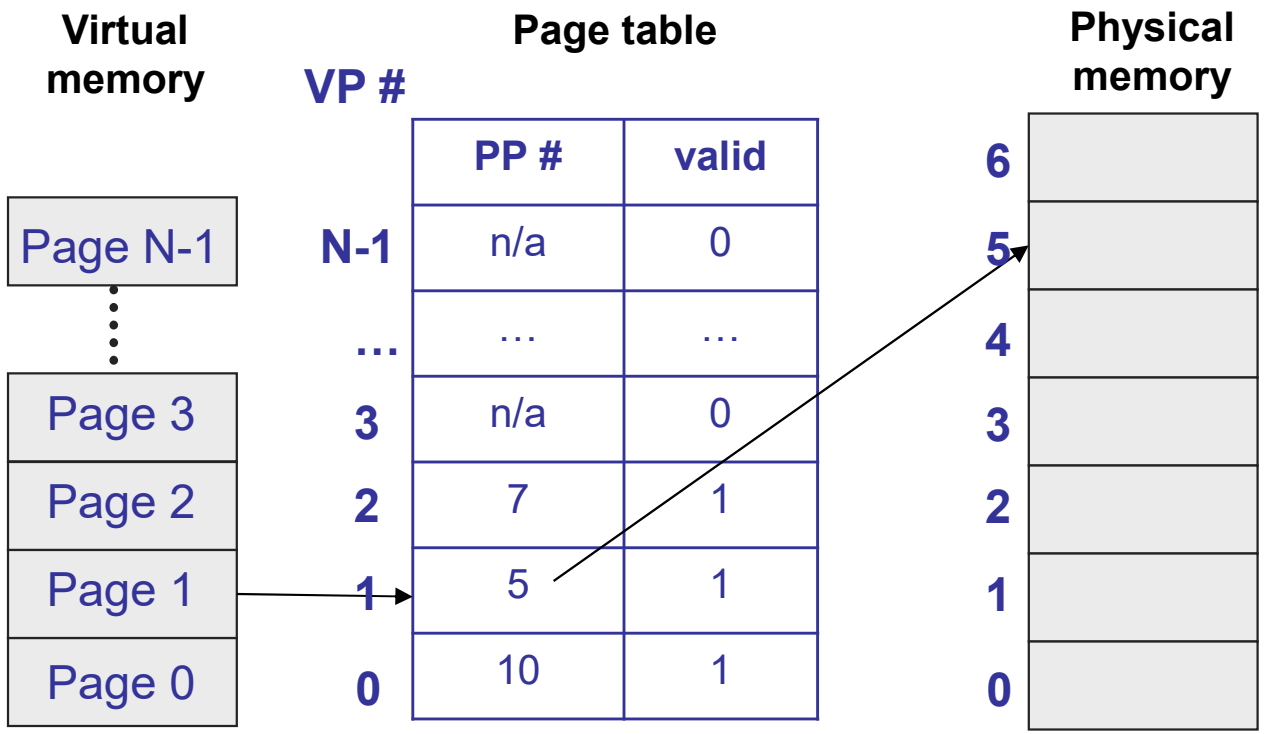
- Virtual address is of the form: (page ~~segment~~ #, offset)

Example (32-bit address, 4096 byte page)

Page table

Virtual page #	Physical page #	Valid
1048575	n/a	0
…	n/a	0
3	n/a	0
2	7	1
1	5	1
0	10	1

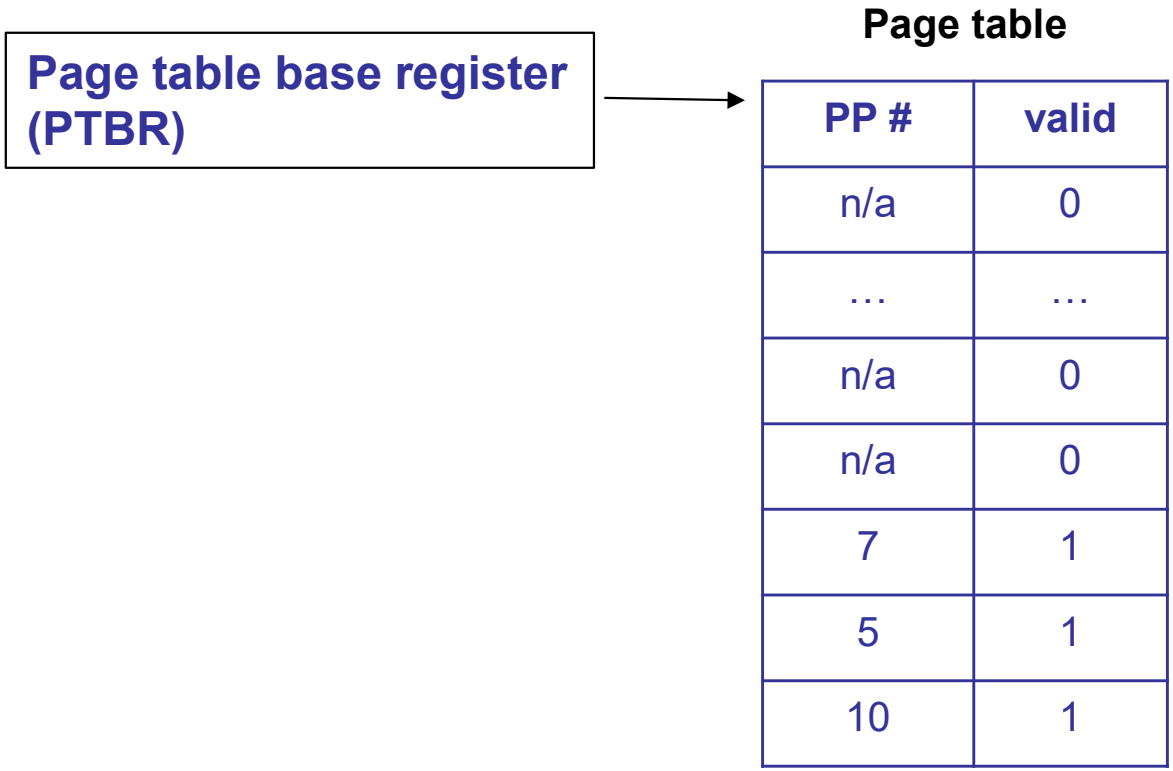
Paging: translation




```
if (virtual page is invalid) {
        trap to OS fault handler
} else {
        physical page # = pageTable[virtual page #].physPageNum
        physical address = {physical page #}{offset}
}
```

Page table

- Lots of entries in page table, so store in memory
  - e.g., 32-bit address space, 4 KB pages ➔ 1M entries in page table
- To change address spaces, must change entire page table!
- How to speed this up?



Page table: additional information

PP #	valid	protection	resident
n/a	0		0
…	…	…	…
n/a	0		0
20	1		1
7	1	rw	1
5	1		0
10	1	r	1

- Each page can be readable or writable (or both or neither)
- Each page can be resident (in memory) or non-resident (on disk)

Paging: MMU algorithm

```
if (virtual page is invalid or protected or non-resident)
        trap to OS fault handler; OS kills process or retries
                                                access
} else {
        physical page # = pageTable[virtual page #].physPageNum
        physical address = {physical page #}{offset}
}
```

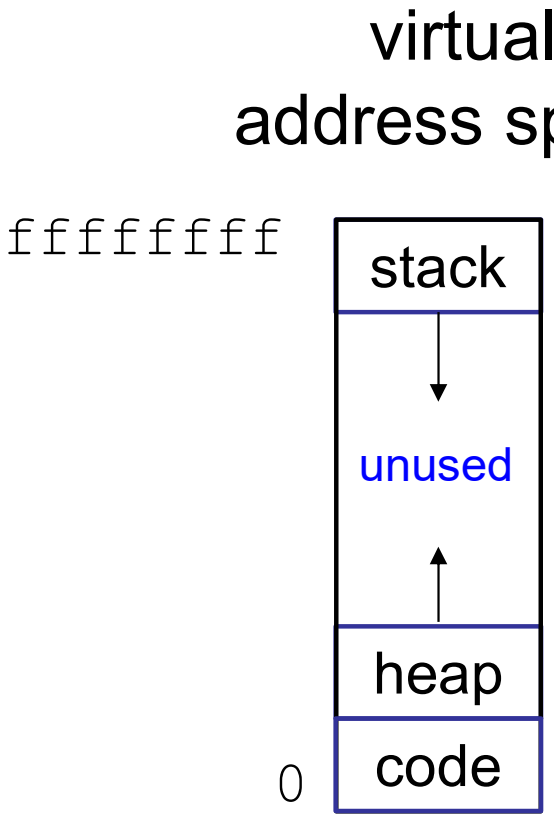
Valid versus resident

PP #	valid	protection	resident
n/a	0		0
…	…	…	…
n/a	0		0
20	1		1
7	1	rw	1
5	1		0
10	1	r	1

- Who makes a virtual page resident/non-resident?
- Who makes a virtual page valid/invalid?
- Why would a process want one of its virtual pages to be invalid?

- Valid → virtual page is legal for process to access. Access to invalid page is an error.
- Resident → virtual page is in physical memory. Access to non-resident page is not an error.

Sparse address spaces



Does this waste space (as it did with base and bound)?

What size should a page be?

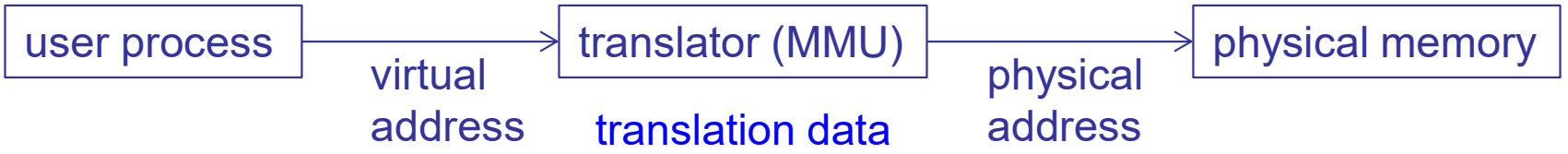
- What happens if page size is really small?
- What happens if page size is really big?

- Typically a compromise, e.g., 4 KB or 8 KB
  - Some architectures support multiple page sizes

Paging

- Pros
  - Easy memory allocation
  - Easy to grow address space
  - Flexible sharing
  - No physical memory used for reserved space in sparse address spaces
- Cons
  - Page table size
  - e.g., 32-bit virtual address, 4 KB pages, 4 byte PTEs
  - 2^{32} bytes of address space ➔ 1M pages ➔ 4 MB page table!
- How to reduce space needed for page table?

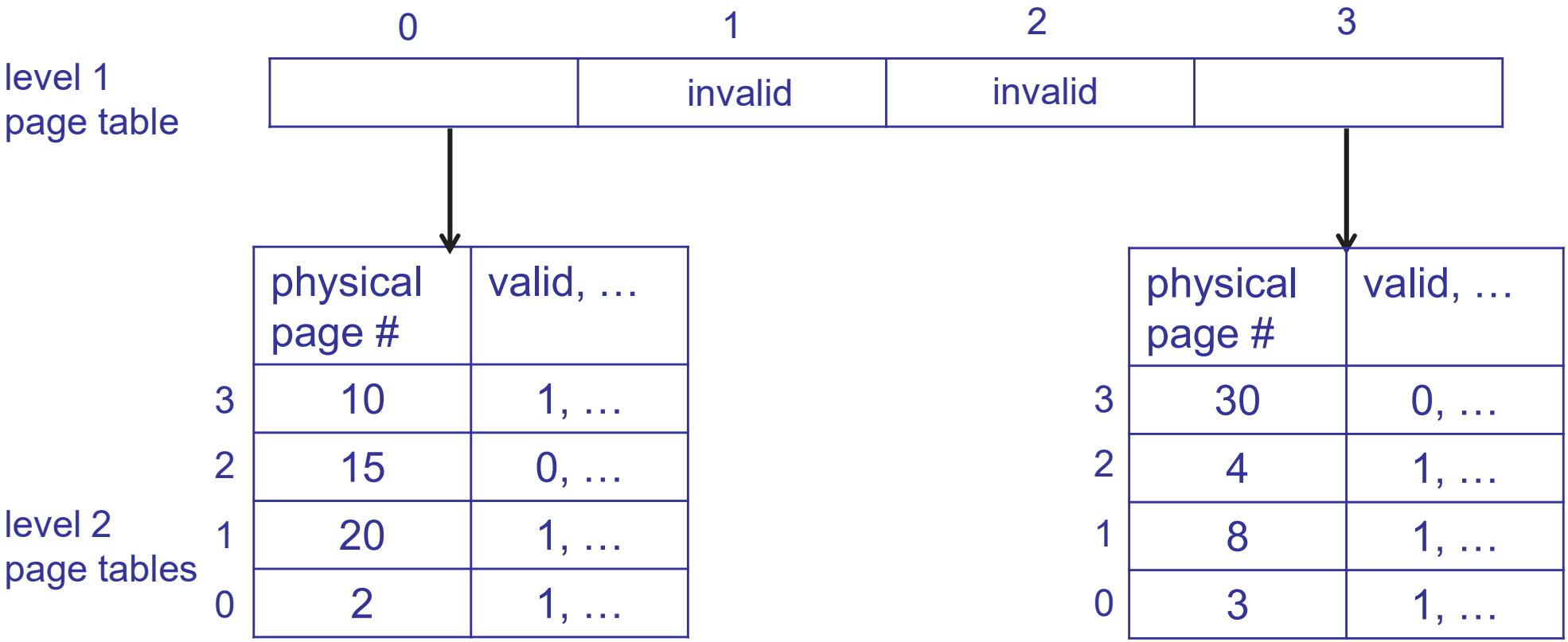
Dynamic address translation



- Many algorithms and data structures can be used to implement translator
  - Base and bound
  - Segmentation
  - Paging
  - Multi-level paging

Multi-level paging

- Standard page table is a simple array
- Multi-level paging generalizes this into a tree



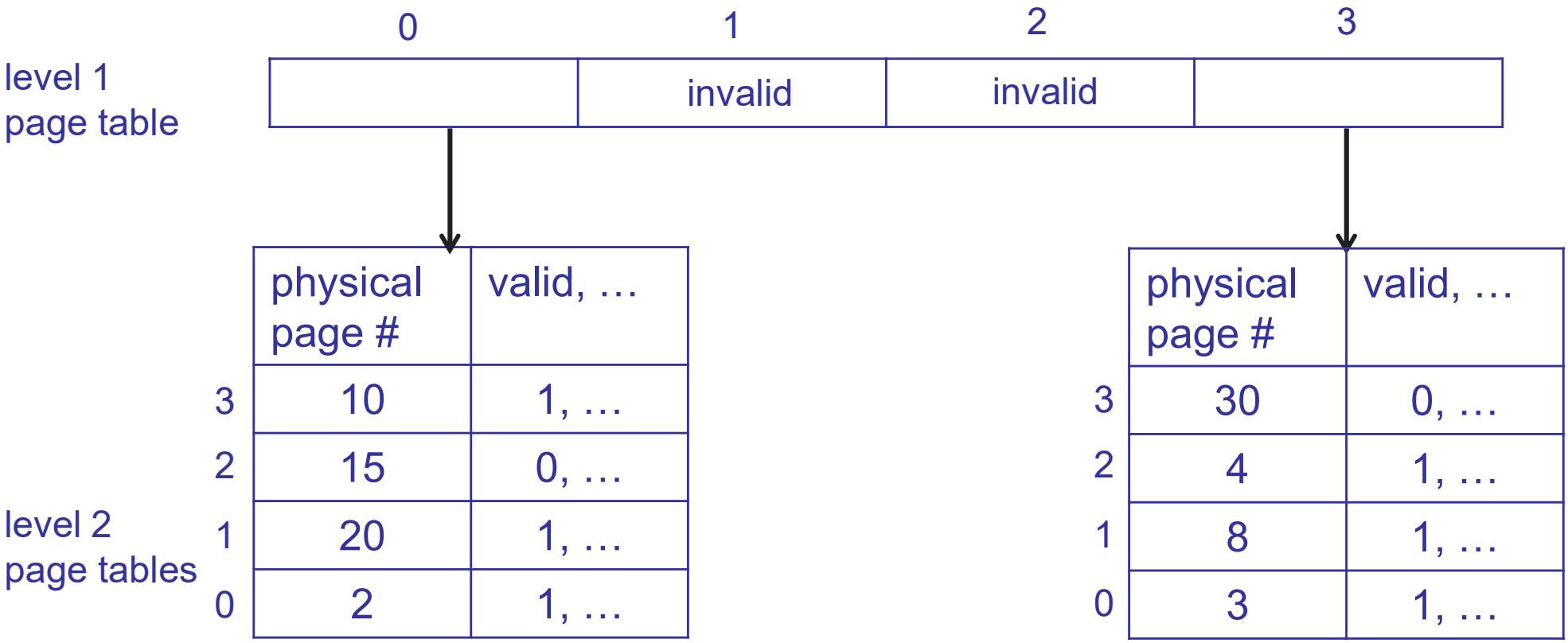
Multi-level paging

Single-level paging

Multi-level paging

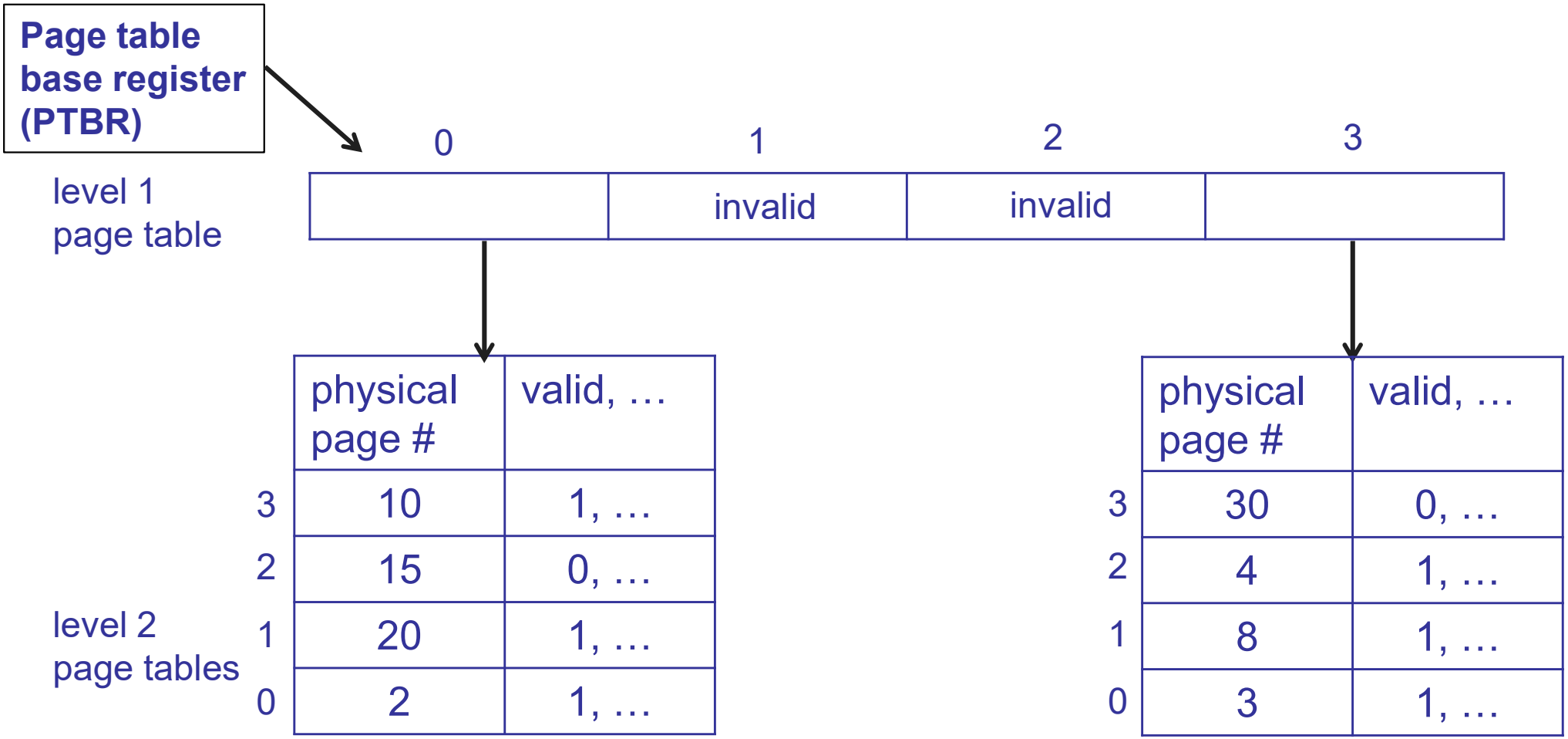
Multi-level paging

- How does this reduce size of translation data?
- Does it always reduce size of translation data?

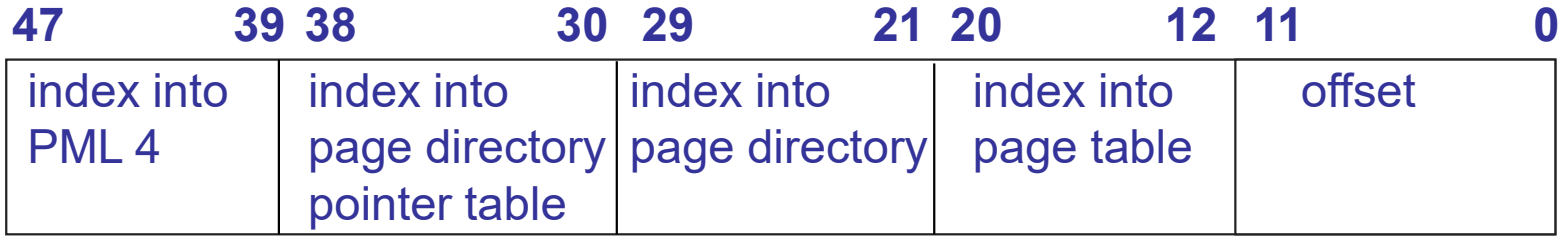


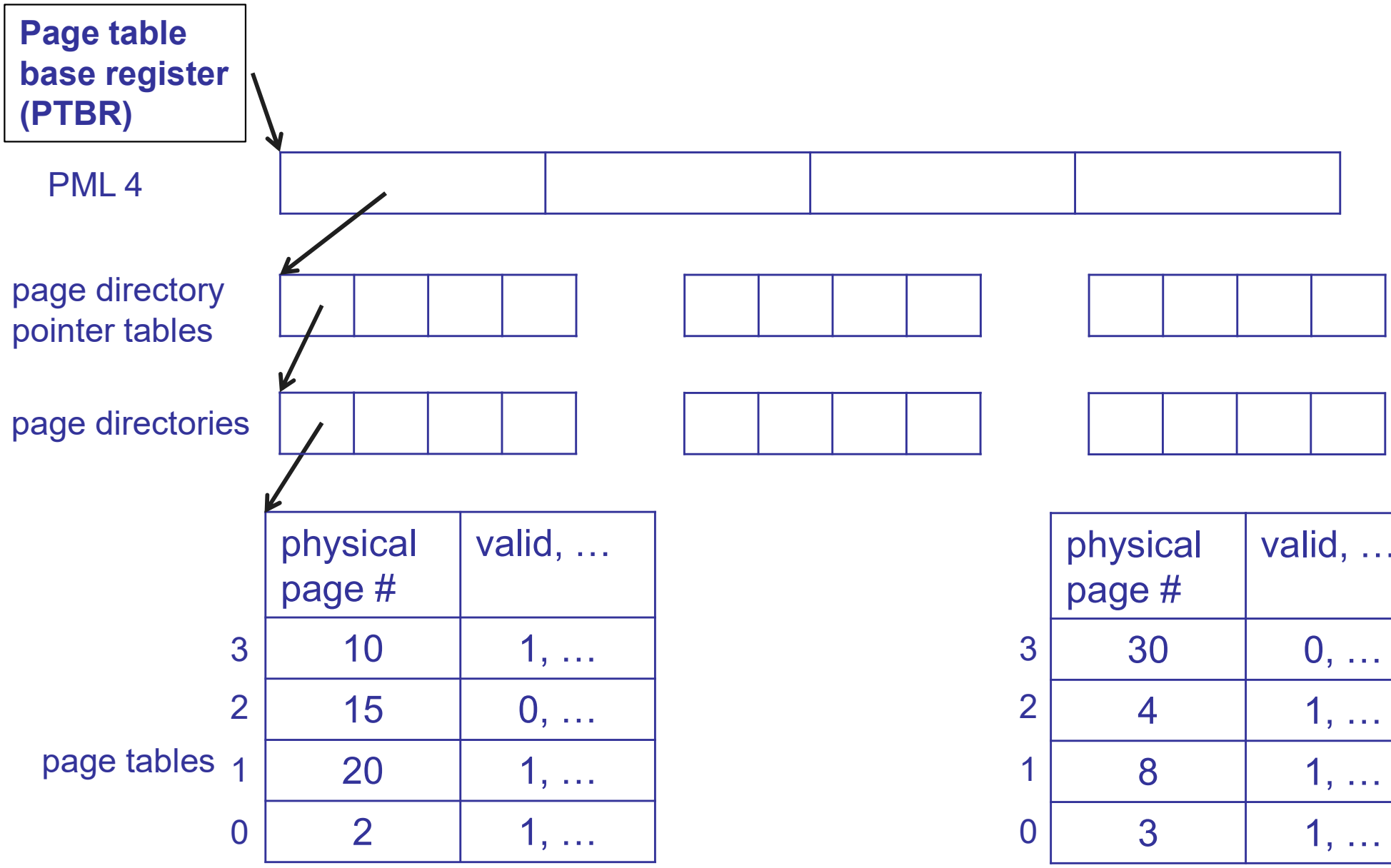
Multi-level paging

- How to share memory between address spaces?
- What must be changed on a context switch?



x86_64


47
39
38
30
29
21
20
12
11
0
index into PML 4
index into page directory pointer table
index into page directory
index into page table
offset


Page table base register (PTBR)
PML 4
page directory pointer tables
page directories
page tables
physical page #
valid, …
3
10
1, …
2
15
0, …
1
20
1, …
0
2
1, …
physical page #
valid, …
3
30
0, …
2
4
1, …
1
8
1, …
0
3
1, …

Multi-level paging

- Pros
  - Easy memory allocation
  - Easy to grow address space
  - Flexible sharing
  - Less translation data used for sparse address spaces
- Cons?
- Solution?

Translation lookaside buffer (TLB)

- TLB: a cache for page table entries
  - TLB hit → Skip all the translation steps
  - TLB miss → Get PTE, store in TLB, restart instruction
    - » x86, ARM: done by hardware
    - » MIPS: done by software

- Must invalidate TLB on context switch, or identify entries with address space ID

Pro tip: indirection + caching

- Indirection: solves all problems in CS!
  - Except performance ☹
- Caching: solves the performance problems caused by indirection
  - Pointers added by indirection are small, so are cached easily